# Exit Ticket · reflect and retrieve

## Exit Ticket

Choosing the cheapest immediate edge always yields the cheapest complete path.

## Use key vocabulary from memory

1. Explain heuristic in your own words.

2. Explain performance model in your own words.

3. Explain pipeline in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## Heuristics, modelling and pipelining

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What does a heuristic provide?

### Q2

What is throughput?

### Q3

What is a pipeline stall?

### Q4

Binary search for 13 in indexes0–14 of [0,1,...,14]. List midpoint indexes.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Define the heuristic — A heuristic quickly estimates how promising a choice is; it is not automatically an exact solution.

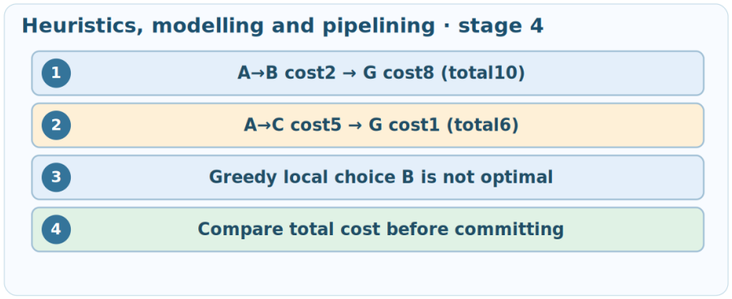
## Core idea 2

Model the trade-off — Separate local edge cost, accumulated cost and estimated remaining cost before choosing a path.

## Core idea 3

Pipeline and performance models — A pipeline can overlap independent stages, increasing throughput, but dependencies and resource constraints may cause stalls.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Evaluate using an A* heuristic and a computational performance model to plan emergency routes. [6]

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Explain why choosing the cheapest immediate edge is not guaranteed to produce a shortest path. [4]

## Question 2

Five independent jobs pass through 3 one-cycle stages. Give ideal completion cycle for all five and compare with non-overlapped 15 cycles. [6]

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

heuristic — Estimate/rule guiding search.

performance model — Simplification estimating resource use.

pipeline — Overlap processing stages.

stall — Delay in progress through stages.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

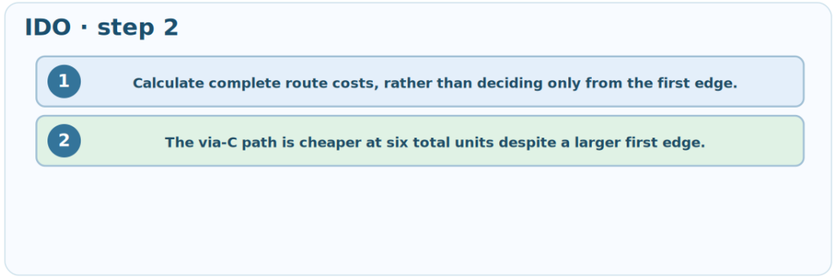
## Problem

Evaluate two routes A–B–G (3+7) and A–C–G (5+1).

## Follow the worked method

1. Predict first — Calculate complete route costs, rather than deciding only from the first edge.
2. Explain the method — The via-C path is cheaper at six total units despite a larger first edge.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict first
Task one occupies S1 in cycle 1, with one new task entering each later cycle.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.